PHYSICAL CHEMISTRY

GUIDED REVISION PLAN-SCORE ADVANCED

Topic : Mole concept & Redox (Question Bank # 05)

MM : 64 **Time : 45 Min.**

SECTION–I : (i) Only One option correct Type

This section contains **9 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

1. 10 ml of 0.10 M solution of a compound ($Na_2CO_3.NaHCO_3 . 2H_2O$) is titrated against 0.05 M HCl. X ml of HCl is used when phenolphthalein is the indicator and Y ml of HCl is used when methyl orange is the indicator in two separate titration, hence (Y – X) is

(A) 20 ml (B) 40 ml (C) 60 ml (D) None of these

2. Calculate the mass of anhydrous oxalic acid, which can be oxidised to CO_2 (g) by 100 ml of an MnO^-_4 solution, 10 ml of which is capable of oxidising 50 ml of $1NI^-$ to I_2.

(A) 45 gm (B) 22.5 gm (C) 30 gm (D) 12.25 gm

3. An aqueous solution contains 2M-$AlCl_3$ and 3m-$SnCl_4$. If the molarity and molality of Cl^- ion in the solution is same, the density of solution is (Al = 27, Sn = 119)

(A) 1.05 gm/ml (B) 1.15 gm/ ml (C) 1.95 gm/ml (D) 2.05 gm/ml

4. A 150 mL of solution of I_2 is divided into two unequal parts. I part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. II part was added with 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2 ?

(A) 0.08 M (B) 0.1 M (C) 0.2 M (D) none of these

5. A 458 g sample containing Mn_3O_4 was dissolved and all manganese was converted to Mn^{2+}. In the presence of fluoride ion, Mn^{2+} is titrated with 3 lit of $KMnO_4$ solution (which was 1.25 N against oxalate in acidic medium), both reactants being converted to a complex of Mn(III), What was the % of Mn_3O_4 in the sample?

(A) 50 % (B) 75% (C) 25% (D) 40%

6. 20 ml of a solution of H_2O_2 is diluted to 100 ml and 50 ml of this solution is titrated with 0.02 M $KMnO_4$ solution in acidic medium and required 50 ml. The volume strength of original H_2O_2 solution is -

(A) 0.285 V (B) 1.12 V (C) 2.80 V (D) 2.28 V

7. A mixture of $NaHC_2O_4$ and $KHC_2O_4 . H_2C_2O_4$ required equal volumes of 0.2 N $KMnO_4$ and 0.12 N NaOH separately. What is the molar ratio of $NaHC_2O_4$ and $KHC_2O_4 . H_2C_2O_4$ in the mixture ?

(A) 6 : 1 (B) 3 : 1 (C) 1 : 3 (D) none

8. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium hydroxide for complete neutralizaton. The percentage of nitrogen in the compound is :

(A) 3% (B) 5% (C) 6% (D) 10%

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9. A mixture of FeO and Fe_2O_3 is reacted with acidified $KMnO_4$ solution having a concentration of 2/5 M, 100 mL of which was used. The solution was then titrated with Zn dust which converted Fe^{3+} of the solution to Fe^{2+}. The Fe^{2+} required 1000 mL of 2/15 M $K_2Cr_2O_7$ solution. Find the ratio by mol of FeO and Fe_2O_3.

(A) 2/3 (B) 3/2 (C) 1/2 (D) 3/1

(ii) One or more options correct Type

This section contains **2 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

10. A definite volume of pure ammonia (NH_3) gas is passed through a series of electric sparks by which the volume becomes 90 ml. The increase in volume is due to formation of nitrogen (N_2) and hydrogen (H_2) gases. All the gases finally present are washed with dilute sulphuric acid solution, by which the volume of gases becomes 80 ml. All the volume are measured at the same temperature and pressure. Which of the following statement(s) is /are correct regarding the original ammonia sample?

(A) The volume of NH_3 gas taken was 40 ml

(B) The volume of NH_3 gas taken was 50 ml

(C) Only 80% of NH_3 gas decomposed into N_2 and H_2 gases.

(D) Only 20% of NH_3 gas decomposed into N_2 and H_2 gases.

11. When $N_2(g)$ and $H_2(g)$ are mixed N_2H_4 (g) , $NH_3(g)$ or both may form , depending upon the relative amount of N_2 and H_2 taken. If initial moles of N_2, H_2 are x, y and final moles of N_2H_4, NH_3 are z, v, then the correct options from the following in order of (x, y , z, v) is/are -

(A) (2, 2, 1, 0) (B) (3, 8, 1, 4) (C) (4,9,4,1) (D) (0.5, 3, 0, 1)

(iii) Paragraph Type

This section contains **1 paragraph** each describing theory, experiment, data etc. **3 questions** relate to 1 paragraphs with 3 questions. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

Paragraph for Q.12 & Q.14

Analysis of dissolved oxygen in river water is very essential to ensure that aquatic life is safe. As per european standards dissolved oxygen should never be less than 6.4 mg / lit and preferably should be greater than 9.6 mg / lit. A water body having value less than 6.4 mg / lit. is considered "heavily polluted" and a water body having value greater than 9.6 mg / lit. is considered optimum fit for aquatic life. One method to measure dissolved oxygen is through Winkler titration. The method involves three stages.

Stage-1 : The water sample is treated with excess of a solution of Mn^{2+} (aq) along with a base to cause the following reaction

$$Mn^{2+}(aq) + 2OH^{-}(aq) \rightarrow Mn(OH)_2(s) \quad \text{... (i)}$$

The precipitate then reacts with dissolved oxygen as shown

$$2Mn(OH)_2(s) + \frac{1}{2}O_2(aq) + H_2O(\ell) \rightarrow 2Mn(OH)_3(s) \quad \text{... (ii)}$$

Stage-2 : The solution is added with excess of acidified solution of KI to cause following reaction,

$$2Mn(OH)_3(s) + 2KI + 3H_2SO_4 \rightarrow 2MnSO_4 + K_2SO_4 + I_2(aq) + 6H_2O \quad \text{... (iii)}$$

Stage-3 : The I_2 produced is calculated by the following reaction by calculating amount of $Na_2S_2O_3$ consumed.

$$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6 \quad \text{... (iv)}$$

Using the above information answer the questions :

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12. If 1 litre of water sample is taken then calculate maximum volume of 0.01 M $Na_2S_2O_3$ solution required if the sample is considered "heavily polluted".

(A) 80 ml (B) 80 litre (C) 120 ml (D) 120 litre

13. If 1 litre of water sample is taken, then calculate minimum volume of 0.1 M $Na_2S_2O_3$ solution required if the sample is considered "optimum fit" for aquatic life.

(A) 8 ml (B) 80 ml (C) 12 ml (D) 120 ml

14. Which of the statements is **correct** w.r.t. above method ?

(A) Greater the amount of dissolved oxygen lesser will be moles of $Na_2S_2O_3$ required.

(B) Excess solution in stage-1 ensures all dissolved oxygen reacts.

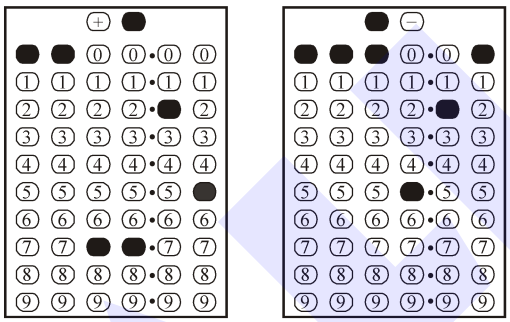
(C) Moles of $Mn(OH)_2(s)$ reacted and moles of $Mn(OH)_3$ formed will be different.

(D) 1 mole of dissolved oxygen will require 2 moles of $Na_2S_2O_3$ for estimation.

SECTION-II: (Integer Second Decimal)

- This section contains **3** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

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1. 2ml Tetrahedral lead, $(C_2H_5)_4$ Pb, is mixed per litre of gasoline to prevent knocking in engine. How many moles of ethyl chloride, C_2H_5Cl, is needed to produce sufficient tetraethyl lead needed to be mixed with 25 L gasoline ? Density of tetraethyl lead is 6.48 gm / ml. (Pb = 208)

$$4C_2H_5Cl + 4Na - Pb \longrightarrow (C_2H_5)_4\,Pb + 4NaCl + 3Pb$$

2. A sample of $K_2SO_4.Al_2(SO_4)_3.24H_2O$ contains 5 kg oxygen then calculate mass of sulphur (in kg) in same sample?

3. Equivalent weight of Br_2 is 96 in the following disproportionation reaction:

$Br_2 + OH^- \rightarrow Br^- + H_2O$ + ? (oxidized product)

The oxidation state of Br in the oxidized product is (Br = 80)

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SECTION–IV : Matrix-Match Type

This Section contains **1 question**. Question has **3 compounds** (A, B and C) given in **Column I** and four values (P, Q, R and S) in **Column II**. Any given statement in Column I can have correct matching with **ONE** or **MORE** statement(s) given in Column II. For example, if for a given question, compounds B matches with the statements given in Q and R, then for the particular question, against value B, darken the bubbles corresponding to Q and R in the ORS.

8(0)

1. To find molarity of 2 separate solutions of $KMnO_4$ and $O_2(O_2 \rightarrow O^{-2})$, a student titrate 20 ml of $KMnO_4$ (in neutral medium) and 10 ml of O_2 separately using 3 reagents A, B & C , performing 3 different experiments for $KMnO_4$ and O_2 each. He found molarity of $KMnO_4$ and O_2 as $\frac{1}{20}$ M and $\frac{1}{10}$ M respectively.

A, B & C are $FeCr_2O_4$, $CuFeS_2$, FeS respectively and their reactions are:

(A) $FeCr_2O_4 \rightarrow K_2CrO_4 + Fe_2O_3$

(B) $CuFeS_2 \rightarrow Cu^+ + Fe^{2+} + SO_2$

(C) $FeS \rightarrow Fe^{+3} + SO_2$

Column-I represents solution of A, B, C used for titration

Column-II represents mili moles of A, B, C consumed with $KMnO_4$ or with O_2.

Column I	**Column II**
(A) $FeCr_2O_4$	(P) $\frac{3}{11}$
(B) $CuFeS_2$	(Q) $\frac{3}{7}$
(C) FeS	(R) $\frac{4}{7}$
	(S) $\frac{4}{11}$

ANSWER KEY

SECTION–I : (i) Only One option correct Type : 3(–1)

1.	**Ans. (B)**	**2.**	**Ans. (B)**	**3.**	**Ans. (D)**	**4.**	**Ans. (B)**
5.	**Ans. (A)**	**6.**	**Ans. (C)**	**7.**	**Ans. (B)**	**8.**	**Ans. (D)**
9.	**Ans. (A)**						

(ii) One or more options correct Type: 4(–1)

10.	**Ans. (B,C)**	**11.**	**Ans. (A,B,D)**

(iii) Paragraph Type: 3(0)

12.	**Ans. (A)**	**13.**	**Ans. (C)**	**14.**	**Ans. (B)**

SECTION-II: (Integer Second Decimal) : 4(0)

1.	**Ans. (4.00)**	**2.**	**Ans. (1.00)**	**3.**	**Ans. (5.00)**

SECTION–IV : Matrix-Match Type : 8(0)

1. **Ans. (A) → Q, R ; (B) → P,S ; (C) → Q, R**

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